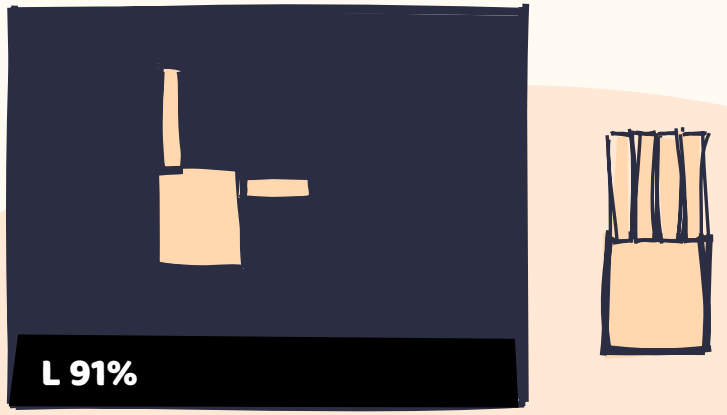
Young Innovators · Class 31

Sign Language Alphabet Recogniser

Teach a computer to read hand signs, and think about who it's for.

Classifier · Fair data · p5.js app



Did you know?
Pakistan Sign Language (PSL) is a full language with its own grammar, used by Deaf communities across Pakistan.

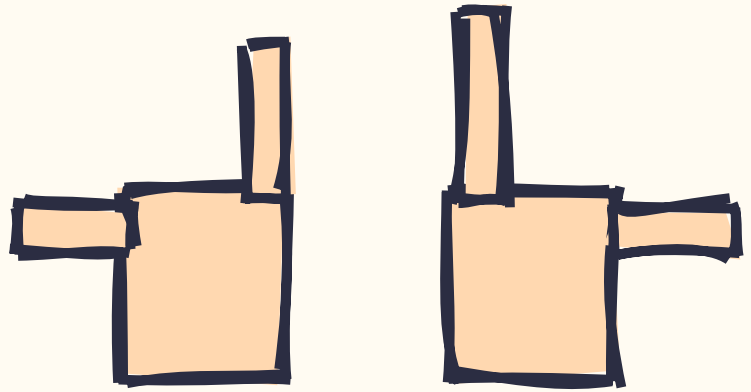
LEARN

Let's think...

Can a webcam understand hands?

Talking with hands

Your classmate signs a letter. You might not understand, but could a laptop camera learn to? And if it could, who should decide how it's used: the programmers, or the people who actually sign?



No way!

Only with gloves?

With examples!

Today: you train an AI model to recognise **5 hand signs** and build a live **p5.js** app that spells words.

Learning objectives

By the end of this class I can...

1	**How it learns:** explain how an **image classifier** learns from labelled examples and gives a **confidence**	**Proof:** I can explain why my model said 'B, 82%'
2	**Train fairly:** train a **Teachable Machine** model for 5 hand signs + 'nothing', with **varied** data	**Proof:** my model works for 3 different classmates' hands
3	**Use it kindly:** run the model live in **p5.js** and discuss how to build it **with** the Deaf community	**Proof:** my app spells a short word, and I can name one limit

LEARN

Objective 3 · Use it kindly · video

Video time!



How AI Can Bridge the Deaf and Hearing Worlds

TED · about 8 minutes · youtu.be/ros3INVOQEU

Watch for…

1. What problem is the speaker trying to **solve**?
2. What can AI do well, and what can it **not** do yet?
3. Why does he say Deaf people must be **involved**?

Scan to watch

After the video: If you built a sign-language app, who would you ask to test it first?

Objective 1 · How it learns

An image classifier learns from examples

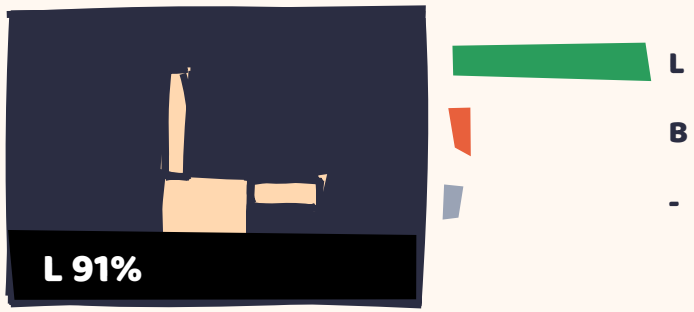
The idea

We give the computer many **labelled** pictures: 'this is L', 'this is B'. It finds patterns in the pixels (edges, shapes) that separate the **classes**.

For a new picture it outputs a **confidence** for each class: L 91%, B 6%, nothing 3%. It picks the highest.

Teachable Machine uses **transfer learning**: a model already trained on millions of images, fine-tuned on your few hundred.

Example: learning to tell twins apart



At first twins look the same. After seeing them many times, you notice small clues: a freckle, a haircut. The model learns small clues in pixels the same way.

Objective 2 · Train fairly

Fair data makes a fair model

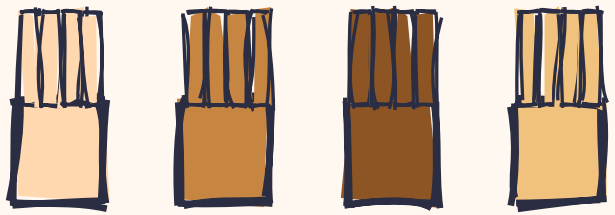
The idea

A model only knows what it was shown. If every example is **one person's hand**, in **one room**, with a **white wall**, it may fail for everyone else. That's **bias**.

Good data: **many people**, different **skin tones**, left and right hands, different **lighting** and backgrounds, near and far.

Always add a **'nothing'** class (no sign) so it doesn't guess letters at random.

Example: learning only from one teacher



If you only ever heard one person speak Urdu, you'd struggle with other accents. Models need variety too.

Objective 3 · Use it kindly

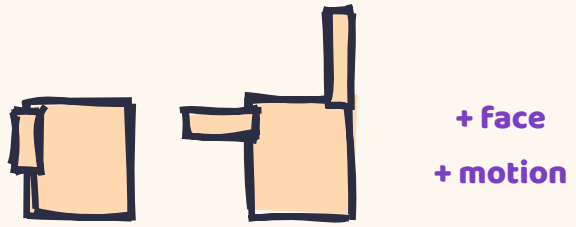
Fingerspelling is not the whole language

The idea

Signing an alphabet letter by letter (**fingerspelling**) is only a tiny part of sign language. Real signing uses **movement**, **both hands**, **facial expression** and grammar. A still-image classifier can't capture that.

Good technology for Deaf people is built **with** Deaf people, not just **for** them. Ask: who needs this? Who tested it? What if it's wrong?

Example: a phrasebook vs a person



A tourist phrasebook helps you say 'hello', but it isn't fluency. An alphabet recogniser is a phrasebook, not an interpreter.

Real world

AI and sign language



PSL resources

Pakistani organisations publish PSL dictionaries and apps



Sign recognition research

Universities train models on video, not single photos



Video relay

Human interpreters on video calls are still the gold standard



Avatars

Some systems turn text into a signing 3D avatar

What could go wrong if a hospital relied only on an AI sign reader?

Nothing about us without us

A motto of disability rights movements: people affected should be part of designing the solution.

Quick quiz

Model thinking

Question	Answer
1. What is a 'class' in our model?	
2. Model says L 55%, B 40%. Trust it?	
3. Why add a 'nothing' class?	
4. All samples from one hand. Risk?	
5. Can our model read full PSL sentences? Why?	

Answers

Cover! 1: one label, e.g. 'L' · 2: not very: it's unsure · 3: so it doesn't guess when no sign is shown · 4: bias, fails for others · 5: no: real signing uses motion, faces, grammar

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human classifier
2. Our AI lab
3. Train the model fairly
4. Live app in p5.js
5. Test for bias and spell words
6. Bias log
7. Responsible AI challenges

MAKE

Class game

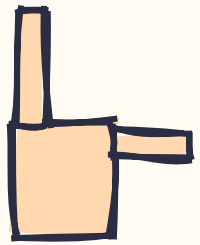
Human classifier

How to play

1. Teacher shows 3 example photos each of 3 made-up signs: 'zig', 'zag', 'zoom'.
2. Volunteers are the **model**: they study the examples for 30 s.
3. Teacher shows new photos (different hands, lighting). Model shouts a label AND a confidence: 'zag, 70%!'
4. Then show a photo of NO sign. What does the model say?

Then discuss

Which test photos fooled the human model? What extra examples would help?



Big idea

Models are only as good as their examples.

Build it

Our AI lab

Teachable Machine
Image Project → Standard model

Webcam
Laptop camera is fine

Sign chart
5 static letters (e.g. A, B, L, V, Y)

editor.p5js.org
For the live app

3+ volunteers
Different hands for fair data

Bias log
Who and what we tested

Choose 5 letters that are held still (no movement). Check them on a PSL or ASL chart with your teacher.

Privacy
Teachable Machine trains in your browser; images aren't uploaded unless you save the project. Only record people who agree.

MAKE

Objective 2 · Build step 1 of 3

Train the model fairly

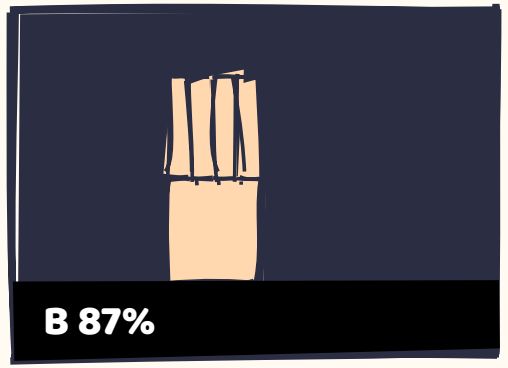
a Create 6 classes: your 5 letters + **nothing** (empty background, random hands).

b Record about **150 samples per class**, from at least **3 people**, both hands, near and far.

c Move around: different backgrounds and lighting.

d Click **Train Model**. Test in the Preview with a person who did NOT give samples.

e Weak letter? Add more varied samples for it and retrain.



Check: Your model recognises all 5 letters for a new person.

MAKE

Objective 3 · Use it kindly · Build step 2 of 3

Live app in p5.js

```
const URL = "https://teachablemachine.withgoogle.com/models/XXXXXXX/"; // yours!
let classifier, video, label = "", conf = 0, word = "";

function preload() { classifier = ml5.imageClassifier(URL + "model.json"); }

function setup() {
  createCanvas(640, 560);
  video = createCapture(VIDEO); video.size(640, 480); video.hide(); classify();
}

function classify() { classifier.classify(video, gotResult); }

function gotResult(err, results) {
  if (err) return console.error(err);
  label = results[0].label;  conf = results[0].confidence;
  classify();                                   // keep going
}

function keyPressed() {
  if (key === " " && label !== "nothing" && conf > 0.8) word += label;
  if (key === "Backspace") word = word.slice(0, -1);
}

function draw() {
  image(video, 0, 0); fill(0); rect(0, 480, 640, 80); fill(255); textSize(28);
  text(label + "  " + nf(conf * 100, 0, 0) + "%", 15, 515);
  text("Word: " + word, 15, 548);
}
```

ml5

The Teachable Machine p5.js export already adds the ml5 library to index.html.

Spell

Hold a sign, press Space to add the letter. Only if confidence > 80%!

Feedback

Showing the % helps the user see when the model is unsure.

Objective 3 · Build step 3 of 3

Test for bias and spell words

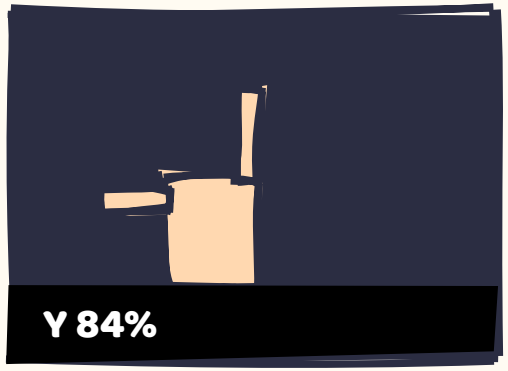
a Each group member tests every letter 5 times. Record correct / wrong in the bias log.

b Test in a **darker corner** and with a **busy background**. What changes?

c Spell 3 short words with your letters (e.g. BAY, LAB if your letters allow).

d Find the letter that fails most. Why? Add data and retrain.

e Write one sentence: who could this help, and what is it NOT good enough for?



Check: Your bias log shows results for 3+ people and 2 conditions.

Test like an engineer

Bias log

Tester / condition	Score	Notes
Person 1 (gave data): correct / 25		
Person 2 (gave data)		
New person (no data)		
Dark room		
Busy background		

Debug checklist

Model never loads? Did you click Upload and copy the link with the final '/'?

Camera blocked? Allow camera in the browser bar.

Always 'nothing'? Hand too far: move closer.

If the new person scores much lower than the people who gave data, your model has learned THEIR hands, not the signs.

MAKE

Level up!

Responsible AI challenges

Choose your challenge

- **More letters:** add 5 more static letters. Which pairs get confused?
- **Speak it:** use p5.speech to say the spelled word out loud.
- **Ask an expert:** invite a PSL user or teacher to test it and give feedback.
- **Hand pose:** try ml5 handPose (21 finger points) instead of whole images.

Think like an engineer

Should a school use this app instead of hiring an interpreter? Why or why not?

Boss level

Collect a much bigger, more varied dataset as a class, with permission, and compare accuracy.

Objective check

Can I really do it? Prove it.

01 Explain how your model decides it's 'B' and what the % means.

02 Show your bias log. Did it work for a new person? What did you change?

03 Demo your app spelling a word, and name one thing it cannot do.

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Ethics circle: each group shares one way their AI could help and one way it could harm, if used carelessly.

Show what you know

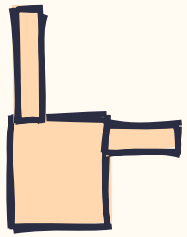
What did we learn?

- [x] Classifiers learn patterns from labelled examples
- [x] Confidence shows how sure the model is
- [x] Varied data from many people reduces bias
- [x] A 'nothing' class stops random guesses
- [x] Build with the people it's meant to help

Exit ticket: why might your model fail for someone who didn't give training samples?

At home

Learn to fingerspell your name in PSL and teach someone at home.



Next class: Build Your Own Game Controller!